KiraOrder

for

ORBYTE

It knows when not to guess — and who to ask.

Falcons · Xiamen University Malaysia · 1 November 2026

The order is not the hard part. The missing facts are.

- Which Gula — the 1 kg pack or the 2 kg pack?
- Which tier price does this customer buy on?
- Is the onion in stock, and how much of it?
- Does “kalau ada” reduce the line, or drop it?
- Does RM2.80 sit below her tier price?
- Which egg grade — A, B or C?

Six facts. The message is synthetic; the six questions are ours. One clerk settles all six today — from memory, at the end of the day.

One clerk. One owner. A phone.

01

One clerk

Takes the order. Settles six facts at once.

02

One owner

Decides the price. Signs the exception.

03

One phone

The message lands here, in three languages.

04

20–80 a day

Orders. Six facts each.

No ERP. No integration. No IT person.

And no one to configure it — so nothing here may ask her to set something up.

All three optimise the message. None can supply a fact the message does not contain.

01

By hand

Ask the customer, read the price file, walk to the whiteboard, type the delivery order.

Nothing is wrong except the time — and it collapses when eleven messages land at once.

02

A translating bot

Malay, Chinese and English become a clean, structured order list.

The list looks perfect and is still wrong. Pack size is not a language problem.

03

A flag-everything checker

Suspicious lines turn red and the flow stops for a human.

A red flag is a question, not an answer. The work still ends with the clerk.

The model proposes. Rules settle. A named person decides.

01 · The model

It proposes. It never prices.

Product candidates, quantities, units, pack sizes and source spans — restricted to the tenant catalogue, or an explicit UNKNOWN.

It cannot set a price, approve an order, or issue a delivery order.

02 · The rules

Facts come from her own data.

Tier price, stock and pack size are computed from the merchant's own catalogue and price file, never generated.

A field the data cannot evidence becomes a blocking issue, never a plausible guess.

03 · The people

A named person signs.

A clerk clears routine orders. The owner decides a below-tier price, and can reject with a reason.

The warehouse sees picking detail with no prices and no contact numbers.

A different shop, the same build.

Changing merchant means changing configuration — seed data, catalogue, aliases, policy. Not code. That is what lets one build sell to a hundred shops.

	SINARAN FOOD SUPPLY (T1)	KILAT HARDWARE & CLEANING (T2)
Category	Food wholesale	Hardware and cleaning supplies
Products	40	16
Inventory	795 units	450 units
Customers	8	5
Application code changed	—	None

Tenant isolation is enforced at the API: a T2 token asking for a T1 delivery order receives 404, with or without a tenant header.

Charged per approved order. Never per seat.

- Billing unit: one approved delivery order. Parsing, fallback, rejection and discard are free — we charge only when the merchant got a delivery order she approved.
- A free tier runs on the rules path: no model call, so a merchant can adopt the discipline of approval and audit before paying anything.
- Measured cost anchor: US$0.00019 per GPU-second on one 24 GB card; at p50 1.93 s per parse that is about US$0.4 per 1,000 orders — model time only, idle GPU excluded.
- The price is not set. Freemium — N free orders a month, then per order — is validated by interview in the pilot. We will not print a number we cannot defend.

Per-order revenue follows work the merchant actually accepts, not seats she hopes to fill.

One order, four layers.

BACKEND & RULES

He Minghao

Where every number comes from — the catalogue, the tier price, the stock — and the approval state machine that refuses a stale edit.

AI & DEPLOYMENT

Xiong Peicheng

What happens when the model fails: the fallback path, the mode on screen, and a timeout derived from a measured p95.

FRONTEND

Ye Lianghui

What a stop looks like: a line that will not settle becomes a card, a question, and a next step the clerk can take.

DATA & QUALITY

Zhou Jingxi

How we score ourselves: a frozen answer key, strict whole-case scoring, and two tenants on one build.

Each layer answers one question the order cannot.

Numbers, failure, interface, evidence — in the order a single order travels.

Where the numbers come from.

INNOVATION

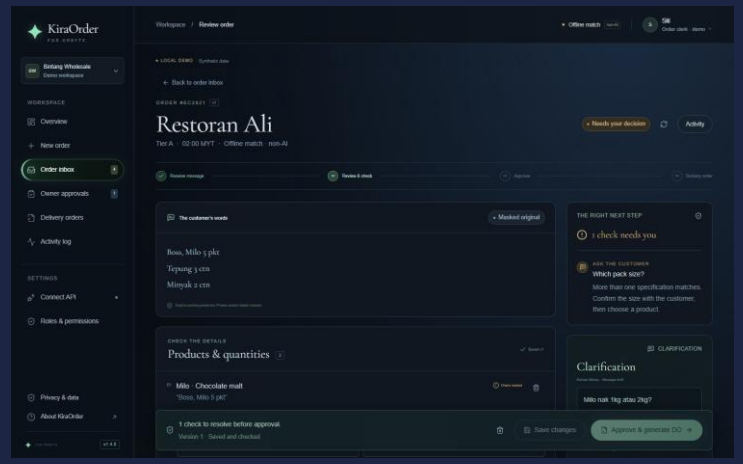
The model never prices.

Every price, every stock quantity and every pack size is computed from the merchant's own catalogue — never generated. A field the data cannot evidence becomes a blocking issue instead of a plausible guess.

CORE TECHNOLOGY

An approval state machine with a permission matrix.

Unsaved edits cannot be approved. A below-tier price cannot pass without the owner. A cross-tenant read returns 404 — enforced at the API, not hidden on the page.



01 · approval blocked, v1.4.0

```
R05 / HE  -  CROSS-TENANT 404 PROOF
========================================================================
time    : 2026-11-01 02:34:52 +08:00
target  : https://kiraorder-api.replit.app/api/v1
session : T2 WAREHOUSE  ->  T1 delivery order
tokens  : read from private JSON; never printed

PS> py -3.13 .\scripts\probe_cross_tenant_404.py --tokens $env:R05_TOKENS

CROSS-TENANT READ PROBE (read-only)
time          : 2026-11-01T02:34:52+08:00
api           : https://kiraorder-api.replit.app/api/v1
object        : DO_20261029_453C4BC2  (tenant T1)
session       : T2 / Ah Kow / WAREHOUSE
authorization : Bearer <redacted: T2_WAREHOUSE>

GET https://kiraorder-api.replit.app/api/v1/warehouse/delivery-orders/DO_20261029_453C4BC2
-> HTTP 404
-> X-Request-ID: 48bc87dc-5ba5-4d0c-9dbb-d52b32c35f39
-> body: {"error_code":"RESOURCE_NOT_FOUND","message":"Resource not found.","details":{}}

control : T1 WAREHOUSE, same object/path -> HTTP 200  [body omitted]

RESULT: PASS - cross-tenant read blocked with 404 RESOURCE_NOT_FOUND

RESULT: PASS  -  T2 session received 404 RESOURCE_NOT_FOUND; T1 control 200.
exit code: 0

Captured for Zhan. Read-only: no write, approve, reset or data change.
```

02 · T2 reads T1 · 404

```
R05 / HE  -  WAREHOUSE PERMISSION MATRIX (C08)
========================================================================
time    : 2026-11-01 02:35:34 +08:00
target  : https://kiraorder-api.replit.app/api/v1
scope   : 41 API permission checks (read/write, roles, tenant, override, method, auth)
tokens  : read from private JSON; never printed

PS> py -3.13 .\scripts\c08_warehouse_audit.py --tokens $env:C08_TOKENS --out .\c08_replay

C08 API matrix: 41/41 passed, field_scan=True, projection_diff=True, status=PASS_API_LAYER
evidence: C:\Users\n_bom\Desktop\r05_he_evidence\c08_replay

RESULT: PASS  -  41 passed / 41 checks, 0 failed.
exit code: 0

Captured for Zhan. Read-only audit: no approve, no order write, no reset.
```

03 · 41 / 41 permission checks

When the model cannot answer, failure is visible.

INNOVATION

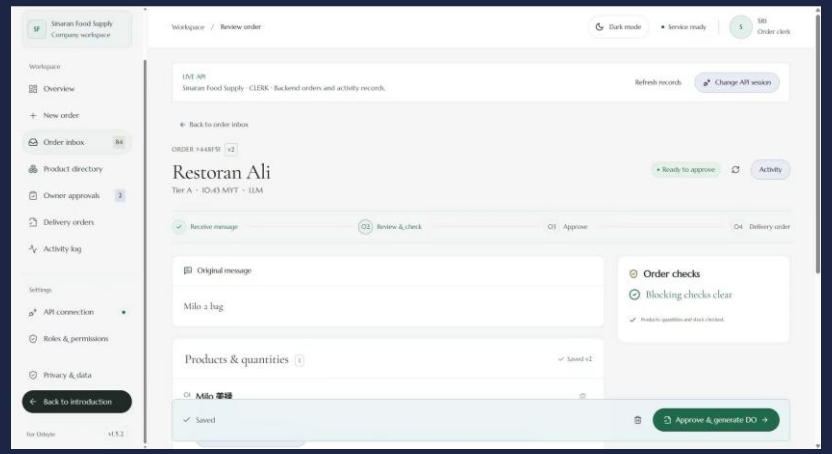
Failure is visible.

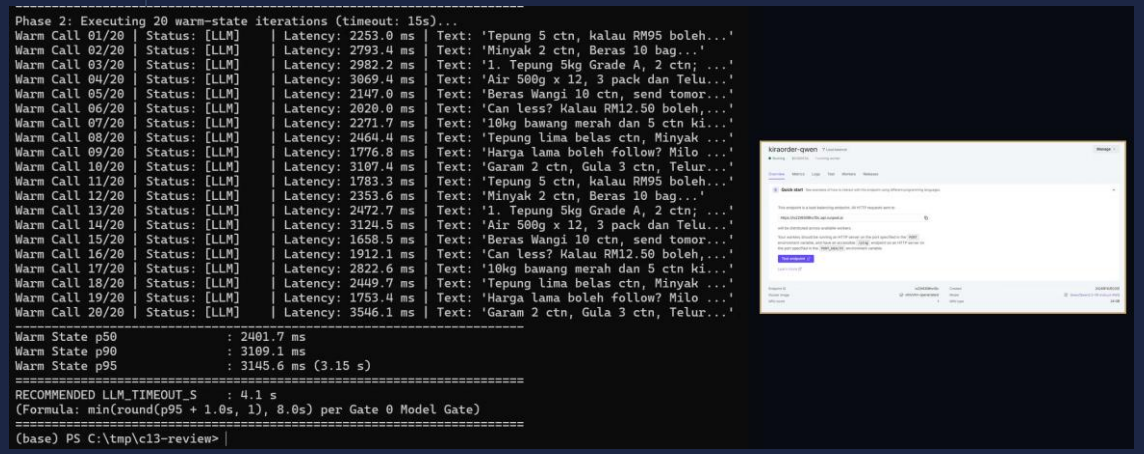
Deterministic fallback, the extraction mode on screen, and no silent guess. If the model is unavailable, the UI shows FALLBACK. If a field is not evidenced, it becomes a blocking issue — not a guess.

CORE TECHNOLOGY

An evidence-bound AI contract.

Qwen proposes product, quantity, unit and pack. The backend verifies each proposed field against the source span; unsupported fields are blocked, not guessed. Price, stock and approval are never decided by the model.





198.5 s
historical true cold start

3.15 s
warm p95 · timeout 4.1 = round(p95 + 1 s)

20 · 0
warm calls · fallbacks — reliability, not accuracy

Make uncertainty actionable.

INNOVATION

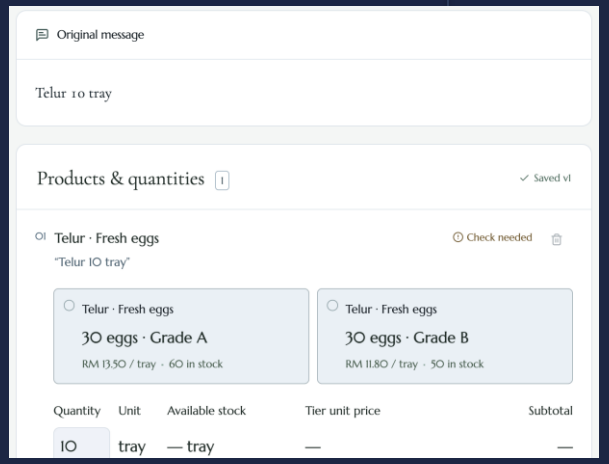
The screen shows both options.

Ten trays of eggs, no grade. Instead of choosing for the clerk, the card puts the two catalogue options side by side — each with its own price and its own stock.

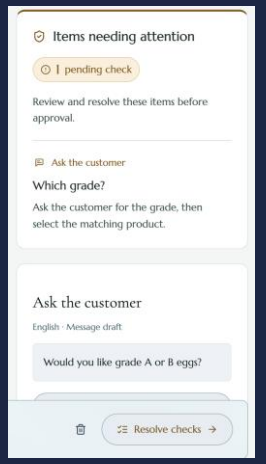
CORE TECHNOLOGY

A dispute becomes a card and a question.

The unresolved line names the field, keeps the span it came from, and drafts the question for the customer. The next step is “Resolve checks”. Approval must wait.

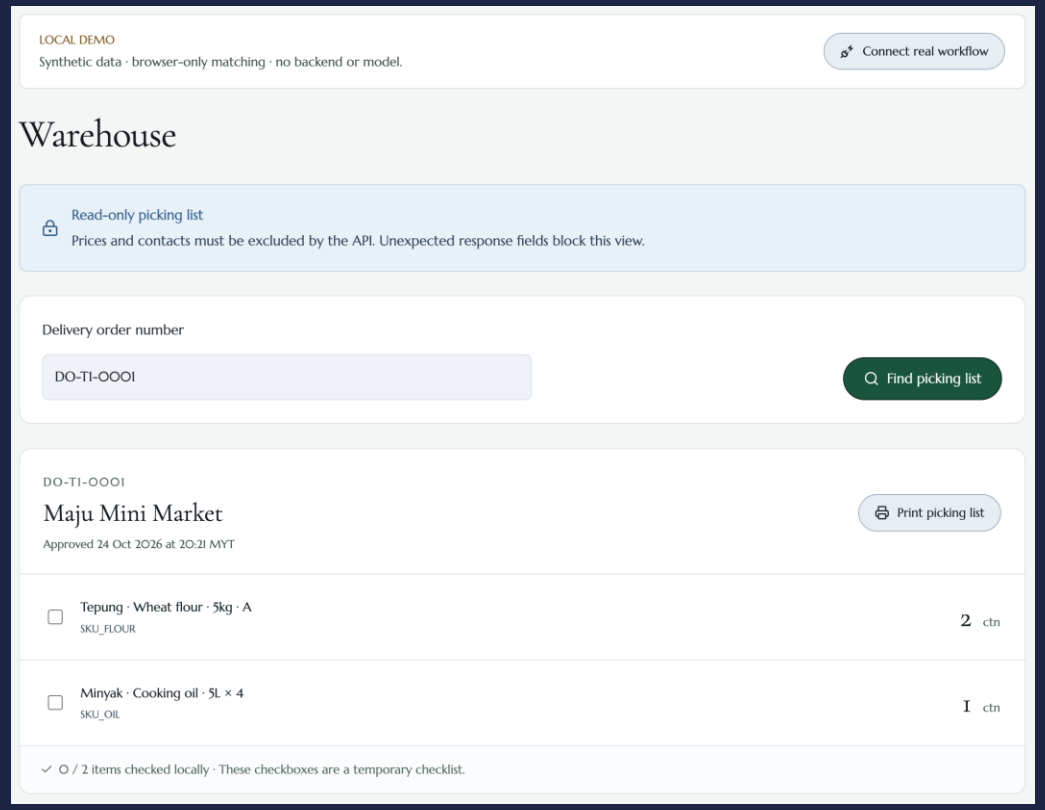


01 · a line that will not settle



02 · the question, drafted

Each role gets only what it needs for the next step.



03 · warehouse, four fields

No prices, no totals, no customer contacts — enforced by the API, not hidden on the page.

We score against an answer key the model did not write.

INNOVATION

Strict whole-case scoring.

A case passes only when all six fields are right at once. Eight of twenty-three do. Small sample, built by us — and we report the number as it is.

CORE TECHNOLOGY

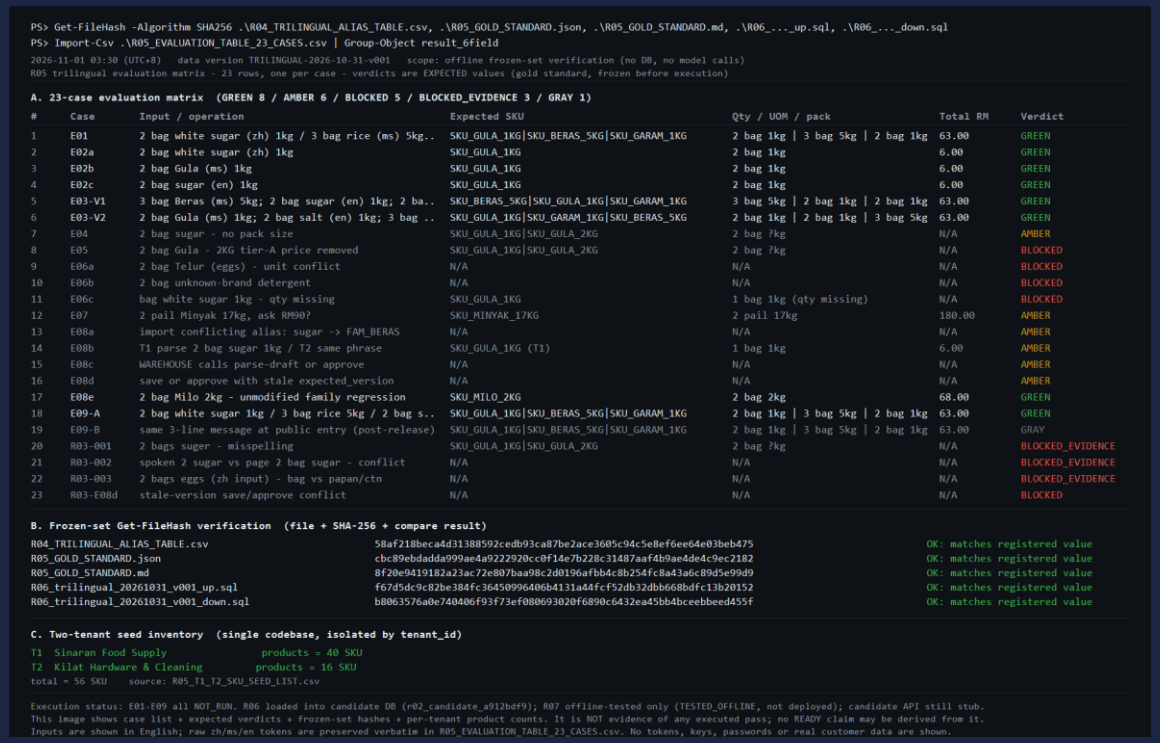
A frozen set with hashes, and two tenants on one build.

Every frozen file carries a SHA-256, so a re-run can be checked rather than trusted. The same build serves a food catalogue and a hardware catalogue.

8/23
all six fields

24/31
correct unit

2
tenants, one build



```
PS> Get-FileHash -Algorithm SHA256 .\R04_TRILINGUAL_ALIAS_TABLE.csv, .\R05_GOLD_STANDARD.json, .\R05_GOLD_STANDARD.md, .\R06_..._up.sql, .\R06_..._down.sql
PS> Import-Csv .\R05_EVALUATION_TABLE_23_CASES.csv | Group-Object result_6field
2026-11-01 03:30 (UTC+8)   data version TRILINGUAL-2026-10-31-v001   scope: offline frozen-set verification (no DB, no model calls)
R05 trilingual evaluation matrix - 23 rows, one per case - verdicts are EXPECTED values (gold standard, frozen before execution)

A. 23-case evaluation matrix  (GREEN 8 / AMBER 6 / BLOCKED 5 / BLOCKED_EVIDENCE 3 / GRAY 1)
#   Case      Input / operation                                  Expected SKU                              Qty / UOM / pack                  Total RM  Verdict
1   E01       2 bag white sugar (zh) 1kg / 3 bag rice (ms) 5kg..  SKU_GULA_1KG|SKU_BERAS_5KG|SKU_GARAM_1KG  2 bag 1kg | 3 bag 5kg | 2 bag 1kg  63.00     GREEN
2   E02a      2 bag white sugar (zh) 1kg                         SKU_GULA_1KG                              2 bag 1kg                         6.00      GREEN
3   E02b      2 bag Gula (ms) 1kg                                SKU_GULA_1KG                              2 bag 1kg                         6.00      GREEN
4   E02c      2 bag sugar (en) 1kg                               SKU_GULA_1KG                              2 bag 1kg                         6.00      GREEN
5   E03-V1    3 bag Beras (ms) 5kg; 2 bag sugar (en) 1kg; 2 ba..  SKU_BERAS_5KG|SKU_GULA_1KG|SKU_GARAM_1KG  3 bag 5kg | 2 bag 1kg | 2 bag 1kg  63.00     GREEN
6   E03-V2    2 bag Gula (ms) 1kg; 2 bag salt (en) 1kg; 3 bag ..  SKU_GULA_1KG|SKU_GARAM_1KG|SKU_BERAS_5KG  2 bag 1kg | 2 bag 1kg | 3 bag 5kg  63.00     GREEN
7   E04       2 bag sugar - no pack size                         SKU_GULA_1KG|SKU_GULA_2KG                 2 bag ?kg                         N/A       AMBER
8   E05       2 bag Gula - 2KG tier-A price removed              SKU_GULA_1KG|SKU_GULA_2KG                 2 bag ?kg                         N/A       BLOCKED
9   E06a      2 bag Telur (eggs) - unit conflict                 N/A                                       N/A                               N/A       BLOCKED
10  E06b      2 bag unknown-brand detergent                      N/A                                       N/A                               N/A       BLOCKED
11  E06c      bag white sugar 1kg - qty missing                  SKU_GULA_1KG                              1 bag 1kg (qty missing)           N/A       BLOCKED
12  E07       2 pail Minyak 17kg, ask RM90?                      SKU_MINYAK_17KG                           2 pail 17kg                       180.00    AMBER
13  E08a      import conflicting alias: sugar -> FAM_BERAS       N/A                                       N/A                               N/A       AMBER
14  E08b      T1 parse 2 bag sugar 1kg / T2 same phrase          SKU_GULA_1KG (T1)                         1 bag 1kg                         6.00      AMBER
15  E08c      WAREHOUSE calls parse-draft or approve             N/A                                       N/A                               N/A       AMBER
16  E08d      save or approve with stale expected_version        N/A                                       N/A                               N/A       AMBER
17  E08e      2 bag Milo 2kg - unmodified family regression      SKU_MILO_2KG                              2 bag 2kg                         68.00     GREEN
18  E09-A     2 bag white sugar 1kg / 3 bag rice 5kg / 2 bag s..  SKU_GULA_1KG|SKU_BERAS_5KG|SKU_GARAM_1KG  2 bag 1kg | 3 bag 5kg | 2 bag 1kg  63.00     GREEN
19  E09-B     same 3-line message at public entry (post-release)  SKU_GULA_1KG|SKU_BERAS_5KG|SKU_GARAM_1KG  2 bag 1kg | 3 bag 5kg | 2 bag 1kg  63.00     GRAY
20  R03-001   2 bags suger - misspelling                         SKU_GULA_1KG|SKU_GULA_2KG                 2 bag ?kg                         N/A       BLOCKED_EVIDENCE
21  R03-002   spoken 2 sugar vs page 2 bag sugar - conflict      N/A                                       N/A                               N/A       BLOCKED_EVIDENCE
22  R03-003   2 bags eggs (zh input) - bag vs papan/ctn          N/A                                       N/A                               N/A       BLOCKED_EVIDENCE
23  R03-E08d  stale-version save/approve conflict                N/A                                       N/A                               N/A       BLOCKED

B. Frozen-set Get-FileHash verification  (file + SHA-256 + compare result)
R04_TRILINGUAL_ALIAS_TABLE.csv   58af218beca4d31388592cedb93ca87be2ace3605c94c5e8ef6ee64e03beb475   OK: matches registered value
R05_GOLD_STANDARD.json           cbc89ebdadda999ae4a9222920cc0f14e7b228c31487aaf4b9ae4de4c9ec2182   OK: matches registered value
R05_GOLD_STANDARD.md             8f20e9419182a23ac72e807baa98c2d0196afbb4c8b254fc8a43a6c89d5e99d9   OK: matches registered value
R06_trilingual_20261031_v001_up.sql   f67d5dc9c82be384fc3645099640 6b4131a44fcf52db32dbb668bdfc13b20152   OK: matches registered value
R06_trilingual_20261031_v001_down.sql   b8063576a0e740406f93f73ef080693020f6890c6432ea45bb4bceebbeed455f   OK: matches registered value

C. Two-tenant seed inventory  (single codebase, isolated by tenant_id)
T1  Sinaran Food Supply              products = 40 SKU
T2  Kilat Hardware & Cleaning        products = 16 SKU
total = 56 SKU     source: R05_T1_T2_SKU_SEED_LIST.csv

Execution status: E01-E09 all NOT_RUN. R06 loaded into candidate DB (r02_candidate_a912bdf9); R07 offline-tested only (TESTED_OFFLINE, not deployed); candidate API still stub.
This image shows case list + expected verdicts + frozen-set hashes + per-tenant product counts. It is NOT evidence of any executed pass; no READY claim may be derived from it.
Inputs are shown in English; raw zh/ms/en tokens are preserved verbatim in R05_EVALUATION_TABLE_23_CASES.csv. No tokens, keys, passwords or real customer data are shown.
```

23 cases frozen before execution: eight expected to pass, eight blocked, six amber, one gray. Every case is still NOT_RUN. Hashes verified; 40 / 16 SKUs on one build.

Homepage, workspace, draft, delivery order.

01 Homepage — dark or light, the merchant's front door

02 Workspace — LIVE API · role: CLERK

03 Restoran Ali · paste 2 bag Gula 1kg · Create draft

04 Read the draft: product, pack size, unit, tier price, stock, source

05 Approve & generate DO — read the number aloud

Point at what it filled — and at what it refused to fill.

It stops — and it stops at the right desk.

01 Edit a line but do not save → Approve stays disabled. Unsaved edits cannot be approved.

02 Record a below-tier price on the Gula line → Send to owner → the draft locks while it waits.

03 Owner role: an empty rejection reason is refused → reject with a reason → order closed.

04 Reopen the draft: closed. No approve, no save, no delivery order.

The point is the disabled button and the closed order, not the enabled one.

What just happened is written down. Not everyone sees everything.

01 Draft activity — read the save, deduct and approve timestamps off the screen, as they are

02 Change record — a matching source becomes MANUAL_SELECTED; pack size 1 kg

03 Warehouse role — find the DO by number: four fields, no price, no total, no contact detail

Say the actor's name out loud — the log is our answer to "who did this?"

CLOSING

A wrong delivery costs more trust than a slow one.

KiraOrder stops where a guess would cost money — and shows you exactly who it asked.

Zhou Jingxi · He Minghao · Xiong Peicheng · Ye Lianghui · Zhan Zhengrong

Demo hozier123.github.io/sdc-2026-team-project · Code github.com/Hozier123/sdc-2026-team-project